🔢 Arrays

1. Two Sum (LeetCode 1)
2. Best Time to Buy and Sell Stock (LeetCode 121)
3. Merge Sorted Array (LeetCode 88)
4. Contains Duplicate (LeetCode 217)
5. Product of Array Except Self (LeetCode 238)
6. Maximum Subarray – Kadane's Algorithm (LeetCode 53)
7. 3Sum (LeetCode 15)
8. Merge Intervals (LeetCode 56)
9. Container With Most Water (LeetCode 11)
10. Rotate Image 90° (Matrix) (LeetCode 48)
11. Minimum Size Subarray Sum (LeetCode 209)
12. Two Sum II – Sorted Array (LeetCode 167)
13. Rotate Array by K Places
14. Find Missing Number in an Array (LeetCode 268)
15. Move All Zeros to the End
16. Find All Pairs with Given Sum
17. Determine if Subarray with Sum Zero Exists
18. Find Kth Largest Element (LeetCode 215)
19. Find Peak Element (LeetCode 162)
20. Sort Elements at Odd Positions Descending and Even Positions Ascending

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🧵 Strings

1. Longest Substring Without Repeating Characters (LeetCode 3)
2. Valid Parentheses (LeetCode 20)
3. Valid Palindrome (LeetCode 125)
4. Valid Anagram (LeetCode 242)
5. Group Anagrams (LeetCode 49)
6. Longest Palindromic Substring (LeetCode 5)
7. Minimum Window Substring (LeetCode 76)
8. Find Index of First Occurrence in a String (LeetCode 28)
9. String Compression (LeetCode 443)
10. Longest Common Prefix (LeetCode 14)
11. Repeated Substring Pattern (LeetCode 459)
12. Longest Repeating Character Replacement (LeetCode 424)
13. Reverse String In-Place (LeetCode 344)
14. Expand Encoded Strings (e.g., "a1b10" → "abbbbbbbbbb")
15. First Non-Repeating Character in a String

🔗 Linked Lists

1. Reverse Linked List (LeetCode 206)
2. Merge Two Sorted Lists (LeetCode 21)
3. Remove Nth Node from End (LeetCode 19)
4. Detect Cycle – Floyd's Algorithm (LeetCode 141)
5. Add Two Numbers (LeetCode 2)

6. Intersection of Two Linked Lists (LeetCode 160)
7. Palindrome Linked List (LeetCode 234)
8. Linked List Cycle II – Find Start of Cycle (LeetCode 142)
9. Middle of the Linked List (LeetCode 876)
10. Check If Linked List Is Palindromic

🧱 Stacks & Queues

1. Valid Parentheses (LeetCode 20)
2. Implement Queue using Stacks (LeetCode 232)
3. Implement Stack using Queues (LeetCode 225)
4. Min Stack (LeetCode 155)
5. Next Greater Element I (LeetCode 496)
6. Daily Temperatures (LeetCode 739)
7. LRU Cache (LeetCode 146)
8. Stack & Queue via Arrays and Linked Lists
9. Circular Queue Implementation

🌳 Trees

1. Inorder Traversal (LeetCode 94)
2. Validate BST (LeetCode 98)
3. Lowest Common Ancestor of BST (LeetCode 235)
4. Convert Sorted Array to Balanced BST
5. Maximum Depth of Binary Tree

6. Diameter of a Binary Tree
7. Preorder and Postorder Traversals
8. Tree Height and Leaf Count

🔗 Graphs

1. Clone Graph (LeetCode 133)
2. Course Schedule – Topological Sort (LeetCode 207)
3. Number of Islands – DFS/BFS (LeetCode 200)
4. Word Ladder (LeetCode 127)
5. Dijkstra's Shortest Path Algorithm
6. Connected Components in Undirected Graph
7. Graph Representations: Adjacency List/Matrix

📚 Dynamic Programming

1. Climbing Stairs (LeetCode 70)
2. House Robber (LeetCode 198)
3. Coin Change (LeetCode 322)
4. Longest Increasing Subsequence (LeetCode 300)
5. Longest Common Subsequence (LeetCode 1143)
6. Unique Paths (LeetCode 62)
7. Maximum Length of Repeated Subarray (LeetCode 718)
8. Partition Equal Subset Sum (LeetCode 416)

🔁 Recursion & Backtracking

1. Factorial, Fibonacci, Sum of Digits (Basic Recursion)
2. Combination Sum (LeetCode 39)
3. Permutations (LeetCode 46)
4. Subsets (LeetCode 78)
5. Letter Combinations of Phone Number (LeetCode 17)
6. Subsets II (With Duplicates) (LeetCode 90)

🧮 Mathematical Problems

1. Check Even or Odd
2. Count Digits in a Number
3. Reverse a Number
4. Power of a Number
5. GCD (Euclidean Algorithm)
6. Prime Number Check
7. Armstrong Number Check
8. Palindrome Number Check
9. Square Root using Binary Search
10. Print All Divisors of a Number

🔍 Sorting & Searching

1. Binary Search (LeetCode 704)
2. Search in Rotated Sorted Array (LeetCode 33)

3. Search in Rotated Array II (LeetCode 81)
4. Find First & Last Position of Element (LeetCode 34)
5. Search in a 2D Matrix (LeetCode 74)
6. Bubble, Selection, Insertion Sort (Basics)
7. Merge Sort (Implementation)
8. Quick Sort (Implementation)

⚡ Bit Manipulation

1. Single Number (LeetCode 136)
2. Number of 1 Bits (LeetCode 191)
3. Power of Two (LeetCode 231)
4. XOR of an Array
5. Reverse Bits (LeetCode 190)

🧠 Important Algorithms to Know

1. Kadane’s Algorithm – Max Subarray
2. Floyd’s Cycle Detection Algorithm
3. Topological Sort
4. Dijkstra’s Algorithm

✅ 1. Object-Oriented Programming (OOPs)

- Principles: Encapsulation, Inheritance, Polymorphism, Abstraction

- Types of Inheritance (Single, Multiple, Multilevel, Hierarchical, Hybrid)
- Interfaces vs Abstract Classes
- Constructor & Destructor behavior
- Method Overloading vs Overriding
- SOLID Principles (optional but good for high-level roles)

✅ 2. Data Structures & Algorithms (DSA)

- Arrays, Strings (sliding window, two-pointer)
- Linked Lists (Singly, Doubly, Cycle detection)
- Stacks & Queues (Min Stack, LRU Cache, Circular Queue)
- Trees (BST, Traversals, LCA, Balanced Trees)
- Graphs (BFS, DFS, Topological Sort, Dijkstra)
- Hashing, Heaps/Priority Queues
- Recursion & Backtracking (N-Queens, Subsets, Permutations)
- Sorting & Searching (Binary Search, Merge Sort, Quick Sort)
- Dynamic Programming (LIS, Knapsack, LCS, Coin Change)
- Bit Manipulation (XOR, Bitmask problems)

✅ 3. Database Management Systems (DBMS)

- Normalization (1NF, 2NF, 3NF, BCNF)
- SQL Queries (JOINs, GROUP BY, Subqueries, Window Functions)
- Transactions (ACID properties)
- Indexing & Views

- Keys (Primary, Foreign, Candidate)
- ER Diagrams

✅ 4. Operating System (OS)

- Processes vs Threads
- CPU Scheduling Algorithms (FCFS, SJF, Round Robin, Priority)
- Deadlock (conditions, prevention, detection)
- Paging & Segmentation
- Memory Management (Virtual Memory, Thrashing)
- System Calls & Context Switching

✅ 5. Computer Networks (CN)

- OSI & TCP/IP Models
- HTTP/HTTPS, DNS, DHCP, IP Addressing
- TCP vs UDP
- Subnetting Basics
- Network Devices (Router, Switch, Hub)
- Socket Programming Basics (optional)

✅ 6. System Design (for advanced roles)

- Basics of Scalability
- Load Balancing, Caching, Database Sharding

- High-level design of common systems (URL Shortener, Chat App)

✅ 7. Programming Language Concepts

- Your preferred language's syntax (Java, Python, C++)
- Exception Handling
- Memory Management (Heap vs Stack)
- Collections/Containers (like HashMap, ArrayList in Java or dict/list in Python)

✅ 8. Version Control Systems

- Git: `clone`, `commit`, `push`, `pull`, `merge`, `branch`
- GitHub workflows (basic)

✅ 9. Project Discussion

- Explain your projects clearly (Problem → Tech Stack → Features → Challenges)
- Know your GitHub repositories well
- Be ready for deep dives into code or DB schema